

FORM TP 2011144



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MAY/JUNE 2011

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – PAPER 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. The use of silent non-programmable calculators is allowed.

SECTION A

Answer ALL questions in this section. Write your answers in the spaces provided in this booklet.

1. (a) Figure 1 is an incompletely labelled diagram of a prokaryotic cell.

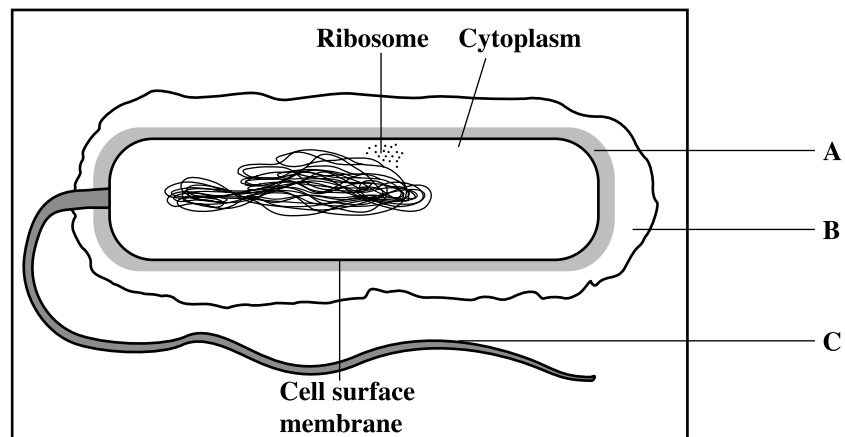


Figure 1. Structure of a prokaryotic cell

- (i) Identify the structures labelled A, B and C in Figure 1.

A _____

B _____

C _____

[3 marks]

- (ii) The structure labelled A in Figure 1 is also present in **some** eukaryotic cells. State how the structure of A in prokaryotic cells differs when present in eukaryotic cells.

[1 mark]

- (iii) There are two types of eukaryotic cells, plant and animal. In which of these is the structure labelled A in Figure 1, NOT found?

[1 mark]

- (iv) Name the structure in a eukaryotic cell that serves the same function as the mesosomes of a prokaryotic cell.

[1 mark]

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- (v) Both eukaryotic cells and prokaryotic cells contain DNA. State ONE way in which the DNA in eukaryotic cells differs from the DNA in prokaryotic cells.

[1 mark]

- (b) Table 1 summarises some of the differences and similarities between typical plant and animal cells. Complete the table by writing the correct answers in the spaces provided.

TABLE 1: COMPARISON OF PLANT AND ANIMAL CELLS

Feature	Plant Cells	Animal Cells
Generation of energy	By mitochondria	
Removal of foreign matter		Digestion by lysosomes
Communication between adjoining cells		Diffusion/ osmosis/ active transport across cell membranes

[3 marks]

- (c) You are provided with a large uncooked potato and six solutions of different concentrations. One of the solutions has the same water potential as the potato.

- (i) Outline a simple investigation to determine which solution has the same water potential as the potato.

[4 marks]

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- (ii) Describe the expected observation in the solution with the same water potential as the potato.

[1 mark]

Total 15 marks

2. (a) In Leghorn chickens, coloured feathers are due to a dominant gene, **F**; white feathers are produced by its recessive allele, **f**. Another dominant gene, **A**, carried on a different chromosome, inhibits colour expression in birds with genotype **FF** or **Ff**. Consequently, both **F_A_** and **ff_ _** are white.

- (i) State the term which describes the interaction of these two genes.

[1 mark]

- (ii) Define the terms 'allele' and 'dominant'.

Allele:

Dominant:

[2 marks]

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- (iii) A coloured rooster is mated with a white hen and produces all coloured offspring.

In the space provided, draw a genetic diagram to explain the cross. **[5 marks]**

(In your diagram, use the symbols **F** to represent the dominant gene for coloured feathers and **f** for the recessive allele which produces white feathers. Use **A** to represent the dominant gene which inhibits colour expression and **a** for the recessive allele.)

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- (b) In pea plants, tall (T) plants are dominant over short (t) plants, and round seeds (R) are dominant over wrinkled seeds (r). A cross between two heterozygous tall, round-seeded plants produced the following progeny:

Tall plants with round seeds	98
Tall plants with wrinkled seeds	46
Short plants with round seeds	44
Short plants with wrinkled seeds	14

- (i) Based on the type of cross, deduce the expected ratio of the phenotypes.

Expected ratio: _____ **[1 mark]**

- (ii) A Chi-square test was performed on the data in (b)(i). The calculated value for χ^2 is 10.81. Using the information provided, formulate a possible null hypothesis for the Chi-square test.

[1 mark]

- (iii) State the formula for, and determine the number of degrees of freedom in the Chi-square test performed in (b)(ii), briefly explaining any terms used.

Formula:

Number of degrees of freedom:

[2 marks]

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- (iv) Use the calculated value of χ^2 (10.81) and the probabilities provided in Table 2 to find the **probability** of the results of the cross departing significantly by chance from the expected ratio. State your findings.

TABLE 2: DISTRIBUTION OF χ^2

Degrees of Freedom	χ^2 values				
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47
Probability (p) that chance alone could produce deviation	0.1 (10%)	0.05 (5%)	0.02 (2%)	0.01 (1%)	0.001 (0.1%)

Probability _____ [1 mark]

- (v) What conclusion may be drawn from the probability found in (b)(iv)?

_____ [2 marks]

Total 15 marks

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3. (a) Figure 2 is a schematic diagram of a human male reproductive system.

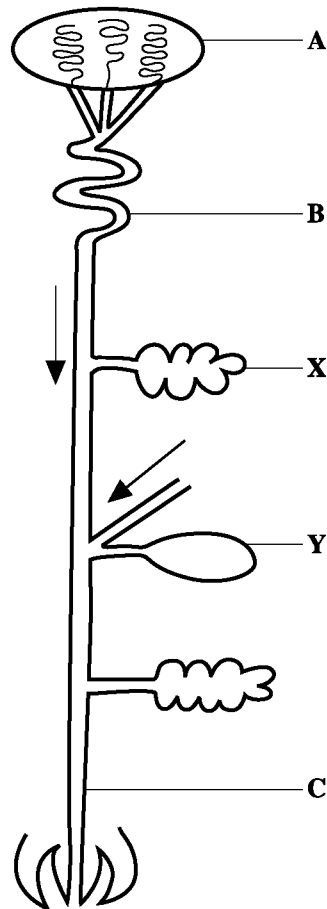


Figure 2. Schematic diagram of a human male reproductive system

- (i) Identify the structures labelled A, B and C in Figure 2.

A _____

B _____

C _____

[3 marks]

- (ii) State the function of EACH of the structures labelled X and Y.

X _____

Y _____

[2 marks]

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- (b) Figure 3 is a drawing of a mature human spermatozoon.

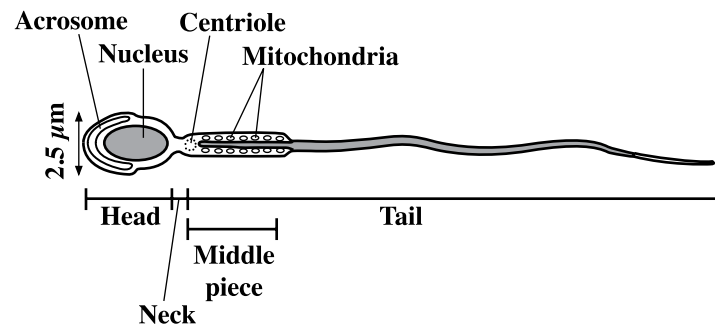


Figure 3. A mature human spermatozoon

- (i) Calculate the actual length of the spermatozoon. Show the steps in your calculation.

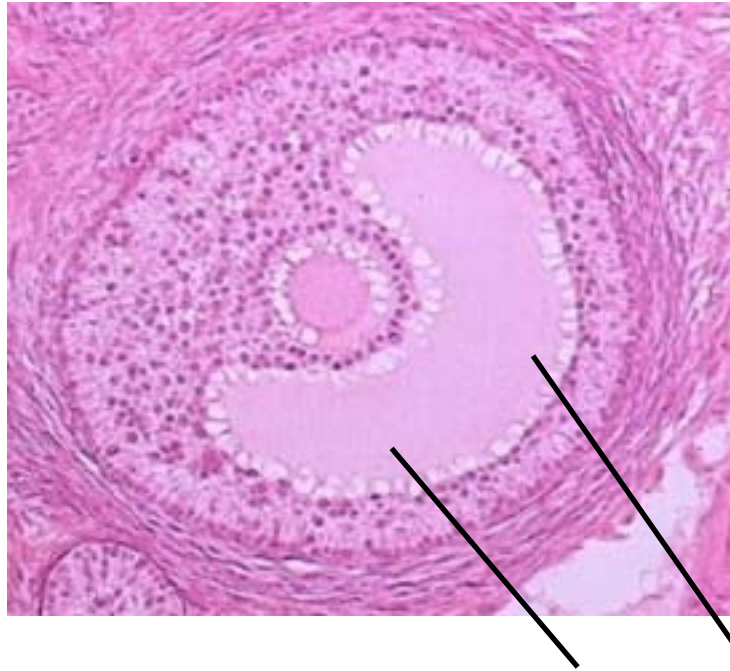
[3 marks]

- (ii) With reference to Figure 3, describe TWO ways in which a mature spermatozoon differs from an ovum.

[2 marks]

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- (c) Figure 4 is a photomicrograph of a section of a mature Graafian follicle.



Source: <http://sprojects.mmi.mcgill.ca/menstrualcycle/Images/>

Figure 4. Mature Graafian follicle

- (i) In the box below, draw and label a plan diagram of the section shown in Figure 4 as identified by the two lines in the lower part of the figure. Make your drawing two times the actual size of the section highlighted.



[4 marks]

- (ii) Name the next stage in the development of the highlighted section in Figure 4.

[1 mark]

Total 15 marks

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5. (a) (i) Outline the stages in Meiosis 1 and comment on the importance of this phase of meiosis. **[6 marks]**
- (ii) State TWO ways in which daughter cells produced by meiosis differ from those resulting from mitosis. **[2 marks]**
- (b) The black billed seed-crackers that inhabit a rainforest in Cameroon, West Africa were found to have EITHER large or small bills, with no birds having bills of intermediate sizes.
- (i) Which of the three types of natural selection BEST describes the example above? Justify your choice. **[3 marks]**
- (ii) Discuss how the frequency of the distribution of phenotypes in the example above would be different if the other two types of selection operated. **[4 marks]**

Total 15 marks

Write your answer to Question 5 here.

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- 6.** (a) (i) Define the term ‘vegetative propagation’. **[2 marks]**
- (ii) Discuss FOUR advantages and THREE disadvantages of using tissue culture as a method of vegetative propagation. **[7 marks]**
- (b) (i) Distinguish between the terms ‘pollination’ and ‘fertilization’. **[2 marks]**
- (ii) Describe TWO reproductive strategies used by plants to promote cross-fertilization. **[4 marks]**

Total 15 marks

Write your answer to Question 6 here.

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